

Matrix functions and stability

Preservation of real H-matrix structure by Newton square-root iterates

Difficulty: challenging **Importance:** interesting to specialist**Status:** Partially resolved

Rating rationale: Challenging because convergence does not control comparison-matrix structure at each iterate; specialist impact concerns square-root algorithms for H-matrices.

Problem statement

For a real square matrix $Y = (y_{ij})$, define its comparison matrix by $M(Y)_{ii} = |y_{ii}|$ and $M(Y)_{ij} = -|y_{ij}|$ for $i \neq j$. Call Y a nonsingular H-matrix when $M(Y) = sI - B$ for some entrywise nonnegative B and real $s > \rho(B)$.

Let $n \geq 1$ and let $A \in \mathbb{R}^{n \times n}$ be a nonsingular H-matrix with $a_{ii} > 0$ for every i . Consider the exact-arithmetic recurrence

$$X_0 = A, \quad X_{k+1} = \frac{1}{2}(X_k + X_k^{-1}A), \quad k \geq 0.$$

Is every X_k a nonsingular H-matrix with strictly positive diagonal? The known existence of these iterates and their convergence to the principal square root do not themselves establish the requested structure at every step.

References and status

N. J. Higham, *Functions of Matrices: Theory and Computation* (SIAM, 2008), §6.3, equation (6.12), §6.8.3, and Research Problem 6.25, p. 170. The author's [errata](#) correct the surrounding p. 161 theorem to real H-matrices; that restriction is adopted here. L. Lin and Z.-Y. Liu, *On the Square Root of an H-matrix with Positive Diagonal Elements*, *Annals of Operations Research* 103 (2001), 339–350, concern the square root itself. Searches on 2026-09-08 for the problem number and Newton/H-matrix structure preservation found no resolution. The recent [Bini–Iannazzo–Meini–Meng preprint](#) (20 May 2026), §§3–4, treats M-matrices, a narrower class. The latest explicit open-problem evidence found for this assertion remains the 2008 book.

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Rechecked the [2026 M-matrix paper](#), §§4 and 4.2: its normalized nonsingular M-matrix class has structure-preserving Newton iterates starting from A . This is a substantive subfamily of the displayed target. Searches for Higham's Problem 6.25 and H-matrix iterate preservation found no general answer; the surviving H-matrix question still has historical evidence.